1. PROBLEM STATEMENT

1.1 Introduction

Voting is an important process not only in democratic systems but also in **organizations such as companies, colleges, and institutions**. It is used for decision-making, leadership selection, and internal elections. Ensuring that voting is conducted in a secure, transparent, and fair manner is essential for maintaining trust among members.

However, traditional voting methods, including paper-based voting and basic electronic systems, suffer from several limitations. These issues reduce reliability, security, and confidence in the voting process.

1.2 Existing System and Its Limitations

Despite technological advancements, existing voting systems still face several challenges:

1.2.1 Lack of Transparency

In traditional systems, vote counting is controlled by a central authority. Members cannot verify whether their votes are correctly recorded, which reduces trust in the results.

1.2.2 Vulnerability to Tampering

Electronic voting systems can be affected by hacking, unauthorized access, and data manipulation. This can lead to incorrect results and loss of system integrity.

1.2.3 Duplicate and Fraudulent Voting

Many systems do not have strong identity verification. This allows individuals to vote multiple times or impersonate others, violating the rule of **“one member, one vote.”**

1.2.4 Centralized Control

Most systems depend on a central authority to manage voting and results. This creates a single point of failure and increases the risk of misuse or manipulation.

1.2.5 Low Trust and Participation

Due to security and transparency issues, members may lose confidence in the system, leading to lower participation in organizational decisions.

1.3 Problem Definition

The main problem addressed in this project is the **lack of a secure, transparent, and tamper-proof voting system for organizations**, which can ensure proper identity verification, prevent duplicate voting, and remove dependence on centralized control.

1.4 Proposed Solution

To overcome these challenges, this project proposes a **Blockchain-Based Organization Voting System with Face Verification**, which includes:

- Face verification for secure employee/member authentication
- Blockchain technology for immutable and transparent vote storage
- Smart contracts to enforce rules like one-person-one-vote

This system ensures that votes are securely stored, cannot be modified, and can be publicly verified within the organization, thereby improving trust and reliability.

1.5 Objectives

- To develop a secure voting system for organizations using blockchain
- To implement biometric (face-based) authentication
- To prevent duplicate and unauthorized voting
- To ensure transparency and auditability
- To reduce dependency on centralized control

1.6 Conclusion

The limitations of traditional voting systems highlight the need for a more secure and transparent solution in organizational environments. By using blockchain and face verification, the proposed system improves security, ensures fairness, and increases trust among members, making it suitable for modern organizational voting needs.

2. LITERATURE REVIEW

2.1 Introduction

Blockchain technology has emerged as a promising solution to enhance the security, transparency, and reliability of electronic voting systems. Researchers have explored various blockchain-based voting models to overcome issues such as vote tampering, lack of trust, and centralized control. This section reviews key research contributions in this domain.

2.2 Detailed Review of Research Papers

1. Blockchain for Electronic Voting System—Review and Open Challenges
https://pmc.ncbi.nlm.nih.gov/articles/PMC8434614/
This paper provides a comprehensive overview of blockchain-based electronic voting systems and highlights their advantages over traditional voting methods. It emphasizes the role of decentralization in eliminating single points of failure. The authors discuss challenges such as scalability, privacy, and usability. The study concludes that blockchain can significantly enhance transparency and trust in voting systems.

2. Blockchain-Based Electronic Voting Systems: A Case Study
https://www.sciencedirect.com/science/article/pii/S2666603024000046
This research presents a real-world implementation of a blockchain-based voting system. It demonstrates how votes can be securely stored as transactions in a

distributed ledger. The system ensures immutability and prevents vote manipulation. The study also highlights the importance of auditability in maintaining election integrity.

3. Blockchain-Based Voting Systems Enhancing Transparency and Security
https://www.itm-conferences.org/articles/itmconf/pdf/2025/07/itmconf_icsice2025_02004.pdf
This paper focuses on improving transparency and security in voting systems using blockchain technology. It explains how cryptographic hashing ensures data integrity. The authors emphasize that decentralized systems reduce dependency on central authorities. The study concludes that blockchain significantly increases voter trust.

4. Decentralized and Automated Online Voting System using Blockchain Technology
https://www.researchgate.net/publication/374505381_Decentralized_and_Automated_Online_Voting_System_using_Blockchain_Technology
This research proposes a decentralized voting system using Ethereum smart contracts. It automates election processes such as vote casting and counting. The system eliminates manual intervention, reducing the chances of human error. The authors highlight improved efficiency and security in the proposed model.

5. E-Voting using Blockchain Technology
https://www.researchgate.net/publication/342932007_E-Voting_using_Blockchain_Technology
This paper discusses the application of blockchain in electronic voting systems. It highlights benefits such as transparency, security, and reduced fraud. However, the study also points out scalability limitations for large-scale elections. The authors suggest further improvements for practical implementation.

6. Blockchain-Based E-Voting Systems: A Systematic Review
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5370817
This study provides a systematic review of blockchain voting systems. It compares different models based on security, scalability, and performance. The authors identify privacy concerns and computational overhead as major challenges. The

paper suggests future research directions for improving blockchain voting systems.

7. Online Voting System using Blockchain
https://www.ijert.org/research/online-voting-system-using-blockchain-IJERTV12IS040091.pdf
This paper proposes an online voting system using blockchain technology. It ensures that votes are stored securely and cannot be altered. The system uses decentralized architecture to eliminate central control. The study concludes that blockchain enhances transparency and reliability.

8. Blockchain-Powered E-Voting: Secure Authentication and Automation
https://indjst.org/articles/blockchain-powered-e-voting-a-novel-approach-to-secure-voter-authentication-online-voting-and-election-automation
This research integrates biometric authentication with blockchain voting systems. It ensures secure voter identity verification and prevents duplicate voting. The use of smart contracts automates the election process. The study highlights improved efficiency and security.

9. Blockchain-Based E-Voting Mechanisms: A Survey
https://www.mdpi.com/2673-8732/4/4/21
This survey analyzes various blockchain-based voting mechanisms. It compares different approaches based on security and performance. The authors discuss cryptographic techniques used to ensure data integrity. The paper concludes that blockchain improves transparency but requires scalability improvements.

10. Digital Voting with Blockchain using IPFS
https://etasr.com/index.php/ETASR/article/view/8440
This paper proposes a voting system combining blockchain and IPFS technology. It ensures secure storage and retrieval of voting data. The decentralized system prevents data manipulation. The authors highlight the importance of distributed storage in enhancing system reliability.

11. Blockchain-Based Electronic Voting System: Significance and Applications
https://onlinelibrary.wiley.com/doi/10.1155/2024/5591147

This study explores the significance of blockchain in modern voting systems. It highlights challenges in identity verification and data security. The authors propose solutions using blockchain technology. The paper emphasizes improved trust and transparency.

12. Hybrid Blockchain-Based Electronic Voting Machine
https://www.mdpi.com/2813-5288/2/4/17
This research proposes a hybrid blockchain-based voting machine. It combines traditional voting methods with blockchain technology. The system ensures security and reliability in vote recording. The authors highlight improvements over conventional EVM systems.

13. Secure and Transparent E-Voting Systems with Blockchain
https://www.mdpi.com/1999-5903/16/11/388
This paper focuses on developing secure and transparent voting systems. It eliminates fraud using blockchain technology. The system ensures that votes are immutable and verifiable. The study highlights improved election integrity.

14. Cybersecurity Issues in Internet and Blockchain Voting
https://academic.oup.com/cybersecurity/article/7/1/tyaa025/6137886
This paper discusses cybersecurity challenges in voting systems. It highlights risks such as hacking and election interference. The authors analyze vulnerabilities in online voting systems. The study emphasizes the need for secure architectures like blockchain.

15. EVM 2.0: Blockchain-Based Voting System
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5241629
This study proposes an improved electronic voting system using blockchain. It enhances security and transparency compared to traditional EVMs. The system ensures accurate vote recording. The authors highlight increased voter trust.

16. Blockchain for Securing Electronic Voting Systems: A Survey
https://link.springer.com/article/10.1007/s10586-024-04709-8
This survey reviews various blockchain-based voting systems. It identifies key benefits such as decentralization and immutability. The authors also discuss

limitations such as scalability. The study provides insights for future improvements.

17. EtherVote: A Blockchain-Based Voting System
https://arxiv.org/abs/2307.10726
This paper introduces EtherVote, a blockchain-based voting platform. It ensures transparency and security in vote recording. The system reduces election costs and improves efficiency. The authors emphasize decentralization benefits.

18. Chirotonia: Secure Blockchain Voting Framework
https://arxiv.org/abs/2111.02257
This research presents a scalable blockchain voting framework. It uses smart contracts for secure vote management. The system ensures data integrity and reliability. The authors highlight improved scalability.

19. Security Requirement Analysis of Blockchain-Based Voting Systems
https://arxiv.org/abs/2208.01277
This paper analyzes security requirements for blockchain voting systems. It discusses key aspects such as privacy, integrity, and authentication. The authors propose solutions to enhance system security. The study highlights the importance of secure design.

20. ElectAnon: Anonymous Blockchain Voting Protocol
https://arxiv.org/abs/2204.00057
This research proposes a privacy-preserving voting protocol. It uses cryptographic techniques to ensure voter anonymity. The system maintains transparency while protecting voter identity. The authors highlight improved privacy in blockchain voting.

2.3 Conclusion

The literature review demonstrates that blockchain-based voting systems provide enhanced security, transparency, and reliability compared to traditional methods. However, challenges such as scalability, privacy, and implementation complexity still need to be addressed. Future research should focus on improving these aspects to enable large-scale adoption.

TITLE:- Secure Blockchain-Based Voting System for Organizations

ABSTRACT

The integrity and transparency of voting systems are critical for ensuring fair elections. However, traditional voting methods, including paper-based systems and electronic voting machines, suffer from several limitations such as lack of transparency, vote tampering, centralized control, and the possibility of duplicate voting. These issues reduce user trust and affect participation in the voting process.

This project proposes a **Blockchain-Based Organization Voting System** to address these challenges. The system utilizes blockchain technology to provide secure, decentralized, and immutable storage of votes. By eliminating the need for a central authority, the system ensures that voting data cannot be altered or manipulated.

The proposed system uses **smart contracts** to enforce voting rules, such as allowing each user to vote only once. Each vote is recorded as a transaction on the blockchain, ensuring transparency and auditability. Additionally, the system includes a public audit feature that allows users to verify voting records and results independently.

The implementation demonstrates that blockchain technology can significantly enhance the security, transparency, and reliability of voting systems. This approach reduces fraud, increases trust, and provides a scalable solution for secure voting within organizations.

LITERATURE REVIEW

Several studies have proposed blockchain-based solutions to improve the security, transparency, and reliability of voting systems. Blockchain technology ensures that data is stored in a decentralized and immutable manner, making it resistant to tampering and unauthorized modifications. These features make it highly suitable for both public and organization-level voting systems.

Jafar et al. (2021) presented a comprehensive review of blockchain-based voting systems, highlighting their ability to enhance transparency and eliminate centralized control. The study also identified key challenges such as scalability and voter privacy, which need to be addressed for large-scale adoption.

Bhavani et al. (2025) explored the integration of blockchain with identity verification mechanisms. Their work emphasized that combining blockchain with secure authentication techniques can significantly improve voter validation and prevent fraudulent activities.

Sujatha et al. (2024) proposed a blockchain-powered voting system using smart contracts. The system automates the voting process, ensuring that votes are securely recorded and counted without manual intervention, thereby reducing errors and improving efficiency.

Sharp et al. (2024) conducted a survey on various blockchain-based voting architectures. The study identified key benefits such as immutability and auditability, while also discussing challenges like system performance and scalability.

Ohize et al. (2025) analyzed different blockchain voting frameworks and evaluated their effectiveness in real-world applications. The study concluded that blockchain technology can significantly improve trust in voting systems.

Chafiq et al. (2024) demonstrated a practical case study of blockchain voting implementation. The results showed that distributed ledger technology enhances auditability and reduces the risk of election fraud.

Somasekhar et al. (2024) proposed a decentralized voting system using distributed ledger technology. The system ensures vote integrity and eliminates reliance on centralized authorities, thereby improving system reliability.

Shaikh et al. (2025) focused on secure voting protocols using blockchain. Their work ensures that each user can vote only once, effectively preventing duplicate voting and maintaining fairness.

Hadiprodjo et al. (2025) studied the efficiency of blockchain-based voting systems. The findings indicate that such systems significantly enhance transparency, security, and reliability.

Benny et al. (2024) designed a decentralized voting framework that reduces dependency on centralized authorities. The system improves overall security and ensures that voting data remains tamper-proof.

IMPLEMENTATION

System Overview

The system is implemented as a blockchain-based voting platform designed for use within an organization. The main objective of this implementation is to provide a secure and transparent way of conducting voting without depending on a centralized system.

In this project, blockchain is used as the core component instead of a traditional backend. All important operations such as storing nominee details, recording votes, and generating results are handled using smart contracts. This ensures that once the data is stored, it cannot be modified.

The system allows an admin to manage the voting process and users to cast their votes using their blockchain wallet. Since all operations are performed on the blockchain, the system becomes more reliable and tamper-proof.

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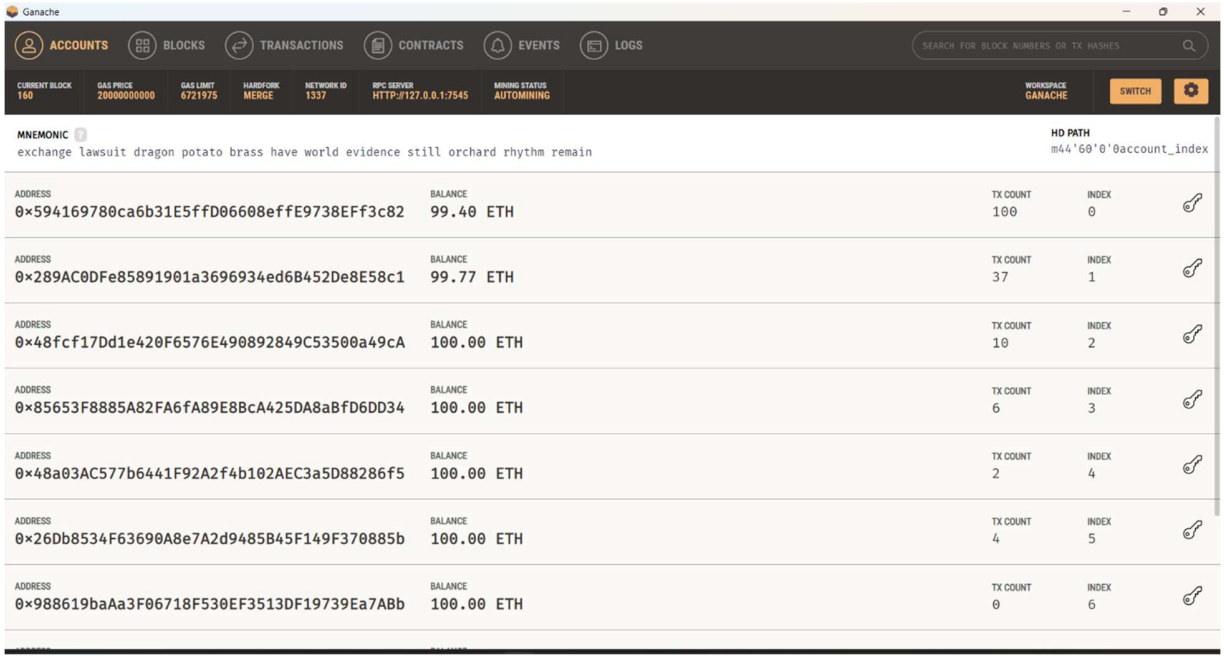
Development Setup

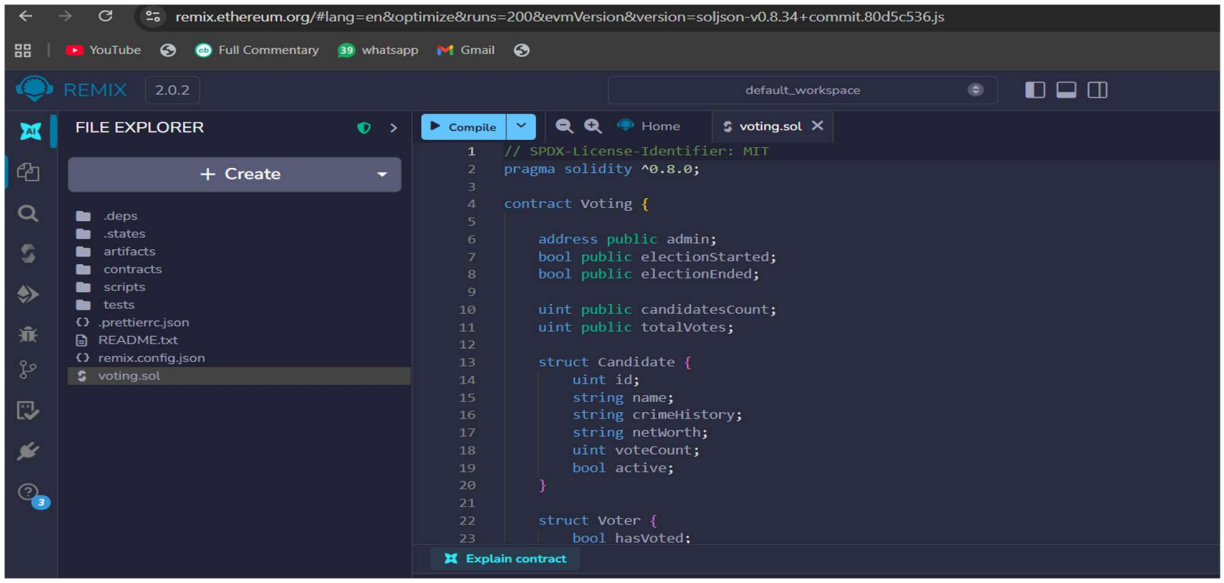
The implementation requires setting up a blockchain environment along with tools that allow interaction with it. For this purpose, Ganache, MetaMask, and Remix IDE are used.

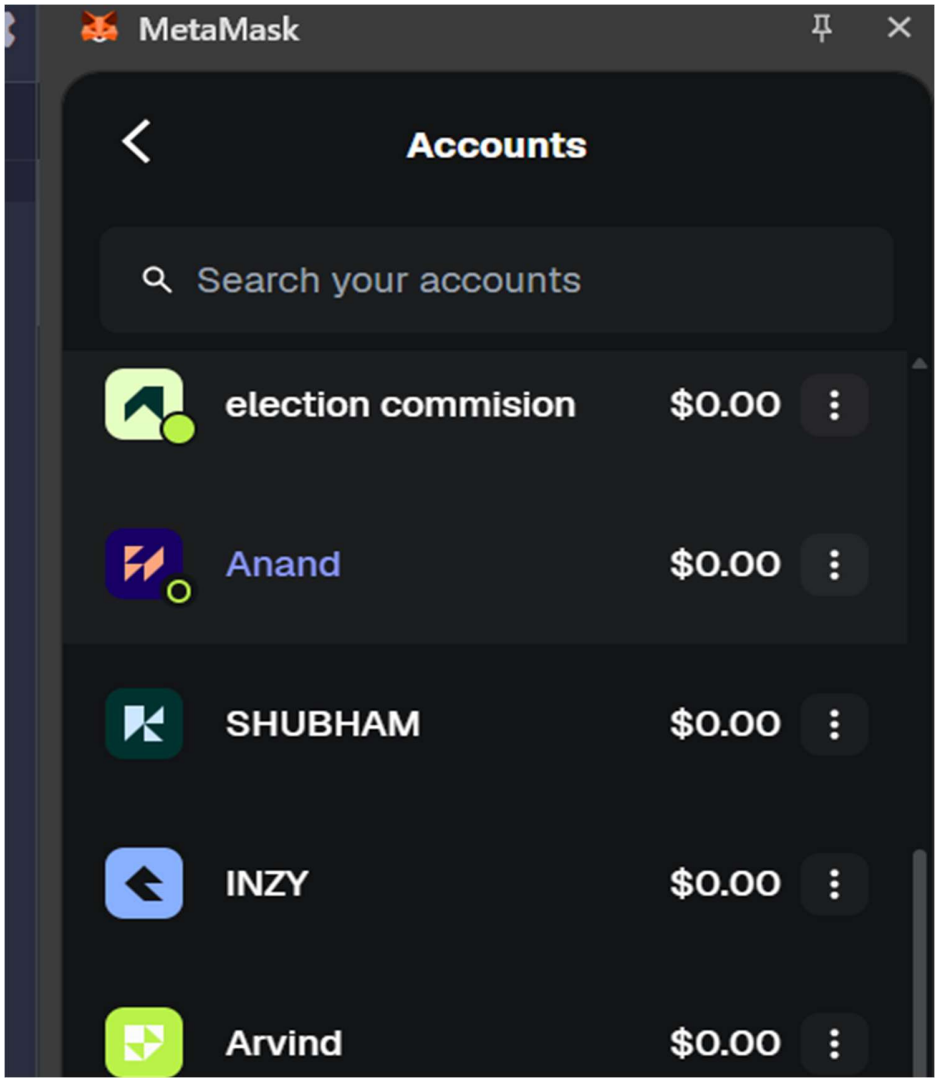
Ganache is used to create a local blockchain network for testing. It provides multiple accounts with test Ether, which are used as admin and voter accounts. All transactions such as adding nominees and voting can be viewed in Ganache.

MetaMask is used as a wallet to connect the application with the blockchain. The Ganache network is configured in MetaMask using custom RPC settings, and accounts are imported using private keys. MetaMask is responsible for signing transactions securely.

Remix IDE is used to write and deploy the smart contract. The contract is compiled and deployed using MetaMask. After deployment, a contract address is generated, which is used in the frontend to interact with the blockchain.


Ganache
ACCOUNTS
BLOCKS
TRANSACTIONS
CONTRACTS
EVENTS
LOGS
SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 160
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 1337
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE GANACHE
SWITCH
MNEMONIC
exchange lawsuit dragon potato brass have world evidence still orchard rhythm remain
HD PATH
m44'60'0'0account_index
ADDRESS 0x594169780ca6b31E5ffD06608effE9738EFf3c82 BALANCE 99.40 ETH TX COUNT 100 INDEX 0
ADDRESS 0x289AC0DFe85891901a3696934ed6B452De8E58c1 BALANCE 99.77 ETH TX COUNT 37 INDEX 1
ADDRESS 0x48fcf17Dd1e420F6576E490892849C53500a49cA BALANCE 100.00 ETH TX COUNT 10 INDEX 2
ADDRESS 0x85653F8885A82FA6fA89E8BcA425DA8aBfD6DD34 BALANCE 100.00 ETH TX COUNT 6 INDEX 3
ADDRESS 0x48a03AC577b6441F92A2f4b102AEC3a5D88286f5 BALANCE 100.00 ETH TX COUNT 2 INDEX 4
ADDRESS 0x26Db8534F63690A8e7A2d9485B45F149F370885b BALANCE 100.00 ETH TX COUNT 4 INDEX 5
ADDRESS 0x988619baAa3F06718F530EF3513DF19739Ea7ABb BALANCE 100.00 ETH TX COUNT 0 INDEX 6


remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js
YouTube
Full Commentary
whatsapp
Gmail
REMIX 2.0.2
default_workspace
FILE EXPLORER
+ Create
.deps
.states
artifacts
contracts
scripts
tests
.prettierrc.json
README.txt
remix.config.json
voting.sol
Compile
Home
voting.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract Voting {
address public admin;
bool public electionStarted;
bool public electionEnded;
uint public candidatesCount;
uint public totalVotes;
struct Candidate {
uint id;
string name;
string crimeHistory;
string netWorth;
uint voteCount;
bool active;
}
struct Voter {
bool hasVoted;
Explain contract



Smart Contract Implementation

The smart contract is the core part of the system, as it controls all voting operations. It is written in Solidity and deployed on the Ethereum blockchain.

In this system, the account that deploys the contract becomes the admin. Only the admin is allowed to add nominees and control the voting process.

Each nominee is stored using a structure that contains details such as name, department, experience, and vote count. These details are stored on the blockchain and cannot be changed after being added.

Mappings are used to store nominee data and to track voters. One mapping is used to store nominee information, and another is used to check whether a user has already voted. This helps in preventing duplicate voting.

The contract includes functions for adding nominees, starting voting, ending voting, and casting votes. The vote function ensures that voting has started and that the user has not already voted before recording the vote.

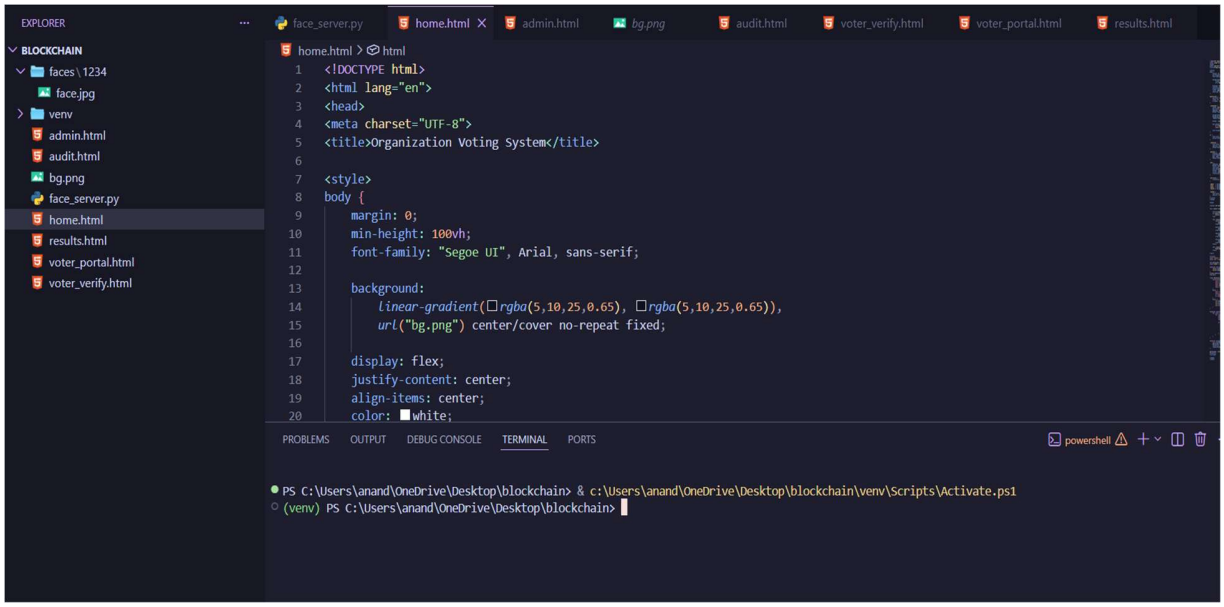
Frontend Implementation

The frontend is developed using HTML, CSS, and JavaScript. It provides an interface for both admin and users to interact with the system.

The home page acts as the main interface and provides navigation to different sections such as admin panel, voting page, audit page, and results page.

In the admin panel, the admin connects their MetaMask wallet. The system checks if the connected account is the admin. Once verified, the admin can add nominees and control the voting process.

The voting page allows users to connect their MetaMask wallet and cast their vote. The list of nominees is displayed, and the user can select one nominee and vote.



Voting Process

The voting process begins when the admin starts the election using the smart contract. Once voting is started, users are allowed to cast their votes.

When a user clicks the vote button, MetaMask opens a confirmation popup. After confirming, the vote is recorded as a transaction on the blockchain.

The system ensures that each user can vote only once by checking whether the wallet address has already been used for voting. If a user tries to vote again, the system prevents it.

Result and Audit

After the voting process is completed, the admin ends the election. Once the election is ended, no more votes can be cast.

The results page displays the number of votes received by each nominee and highlights the winner. Since the data is fetched directly from the blockchain, the results are accurate and cannot be changed.

The audit page provides transparency by displaying blockchain-related information such as admin address, voting status, and transaction logs. This allows users to verify the voting process.

Summary

The implementation of this system demonstrates how blockchain can be used to create a secure and transparent voting platform. By using smart contracts and decentralized storage, the system eliminates the need for a central authority and ensures that the data remains tamper-proof.

Sample Code

REMIX-IDE :

Voting.sol

```
// SPDX-License-Identifier: MIT

pragma solidity ^0.8.0;
```

```
contract Voting {
    address public admin;
    bool public electionStarted;
    bool public electionEnded;
    uint public candidatesCount;
    uint public totalVotes;
    struct Candidate {
        uint id;
        string name;
        string crimeHistory;
        string netWorth;
        uint voteCount;
        bool active;
    }
    struct Voter {
        bool hasVoted;
    }
    mapping(uint => Candidate) public candidates;
    mapping(address => Voter) public voters;
    event ElectionStarted(uint time);
    event ElectionEnded(uint time);
    event CandidateAdded(uint id, string name);
    event CandidateRemoved(uint id);
    event VoteCast(bytes32 voteHash);
    constructor() {
        admin = msg.sender;
```

```
}
modifier onlyAdmin() {
    require(msg.sender == admin, "Only admin");
    _;
}
function startElection() public onlyAdmin {
    require(!electionStarted, "Already started");
    electionStarted = true;
    electionEnded = false;
    emit ElectionStarted(block.timestamp);
}
function endElection() public onlyAdmin {
    require(electionStarted, "Not started");
    electionEnded = true;
    emit ElectionEnded(block.timestamp);
}
function addCandidate(
    string memory _name,
    string memory _crimeHistory,
    string memory _netWorth
) public onlyAdmin {
    require(!electionStarted, "Election already started");
    candidatesCount++;
    candidates[candidatesCount] = Candidate(
        candidatesCount,
        _name,
```

```
        _crimeHistory,
        _netWorth,
        0,
        true
    );
    emit CandidateAdded(candidatesCount, _name);
}
function removeCandidate(uint _id) public onlyAdmin {
    require(!electionStarted, "Election already started");
    require(candidates[_id].active, "Already removed");
    candidates[_id].active = false;
    emit CandidateRemoved(_id);
}
function vote(uint _candidateId) public {
    require(electionStarted, "Election not started");
    require(!electionEnded, "Election ended");
    require(!voters[msg.sender].hasVoted, "Already voted");
    require(candidates[_candidateId].active, "Invalid candidate");

    voters[msg.sender].hasVoted = true;
    candidates[_candidateId].voteCount++;
    totalVotes++;
    bytes32 receipt = keccak256(
        abi.encodePacked(msg.sender, block.timestamp, totalVotes)
    );
    emit VoteCast(receipt);
```

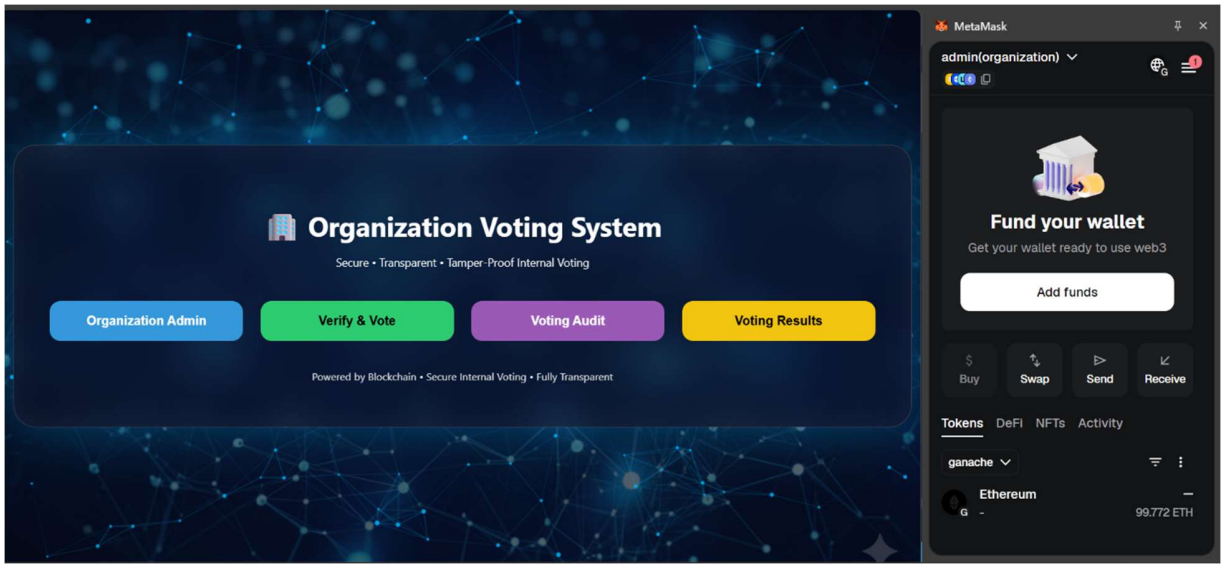
```
    }
    function getElectionStatus() public view returns (string memory) {
        if (!electionStarted) return "Not Started";
        if (electionEnded) return "Ended";
        return "Ongoing";
    }
}
```

Other codes

Git link : https://github.com/amritanand2023/blockchain-voting-system

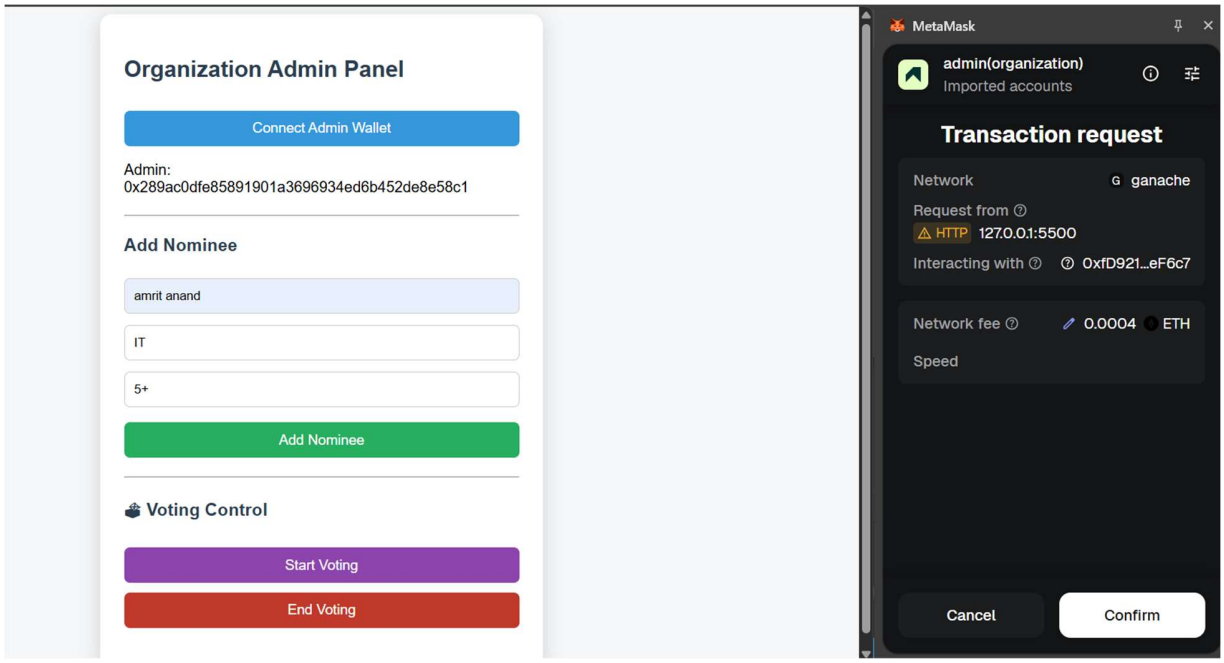
Output Screen shot

Home Page

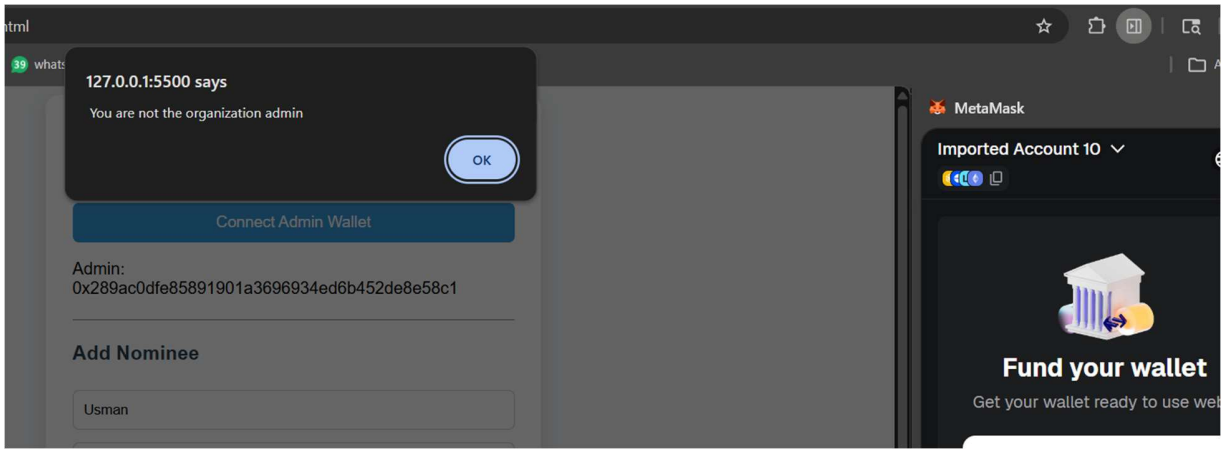


Admin Panel

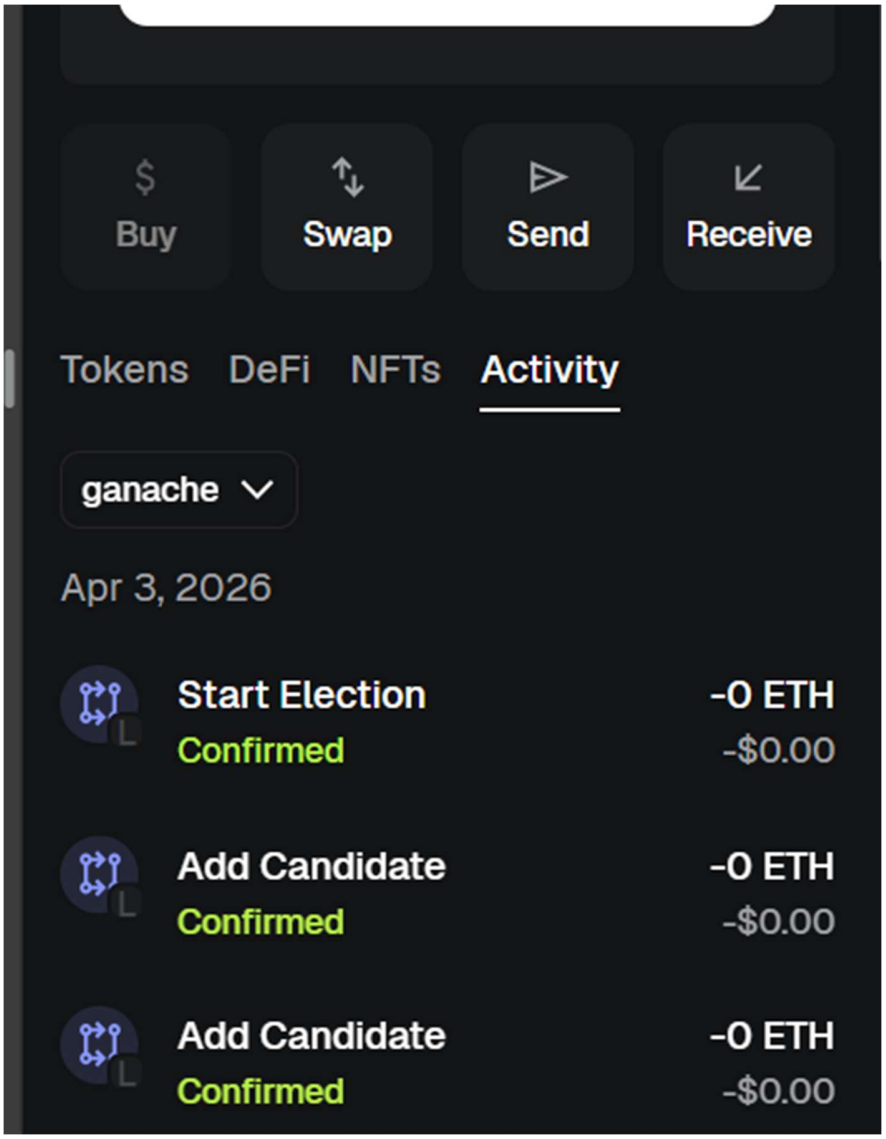
candidates added and voting started



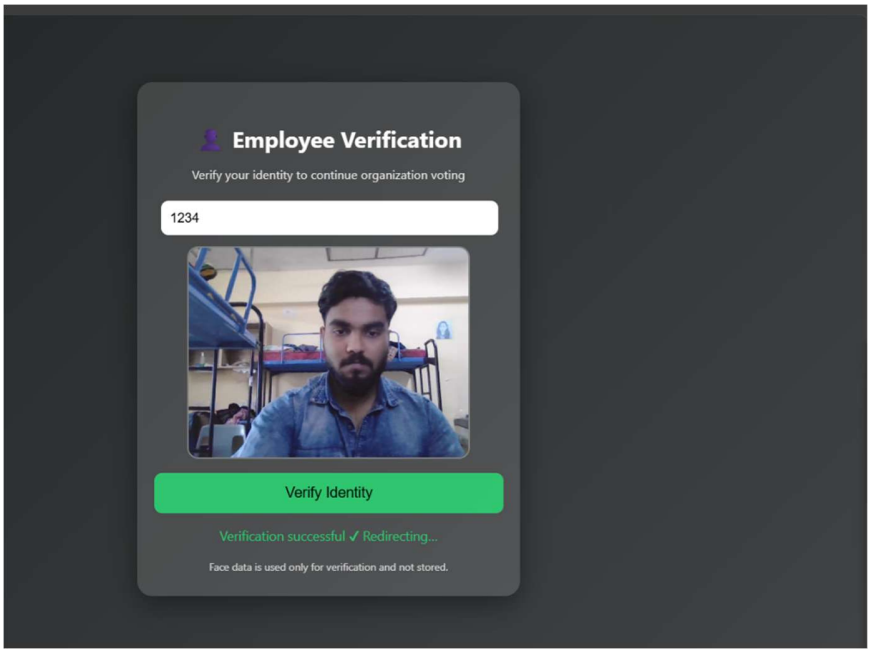
If a non-admin account is used, an alert message is displayed, preventing unauthorized access to admin functionalities.



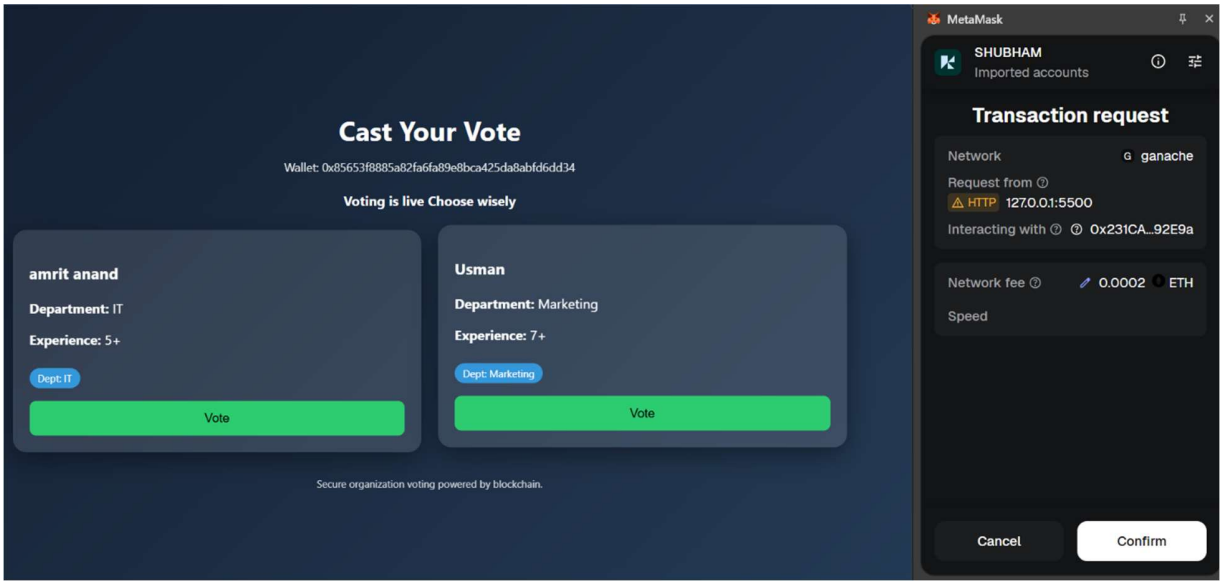
MetaMask Transaction Activity Log (Admin)



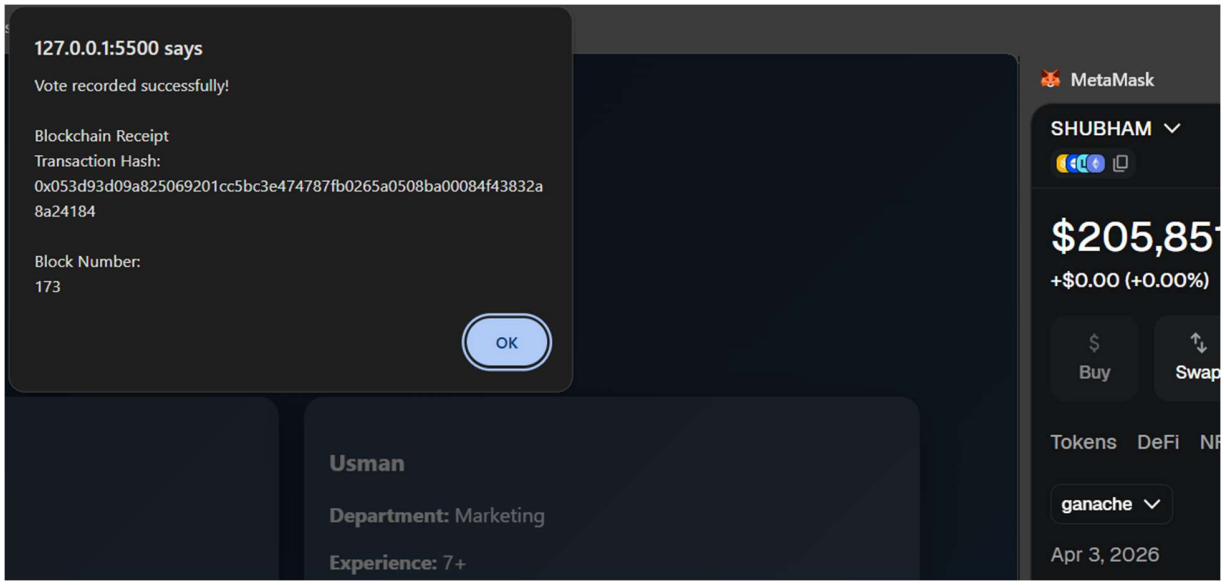
Employee Verification Interface



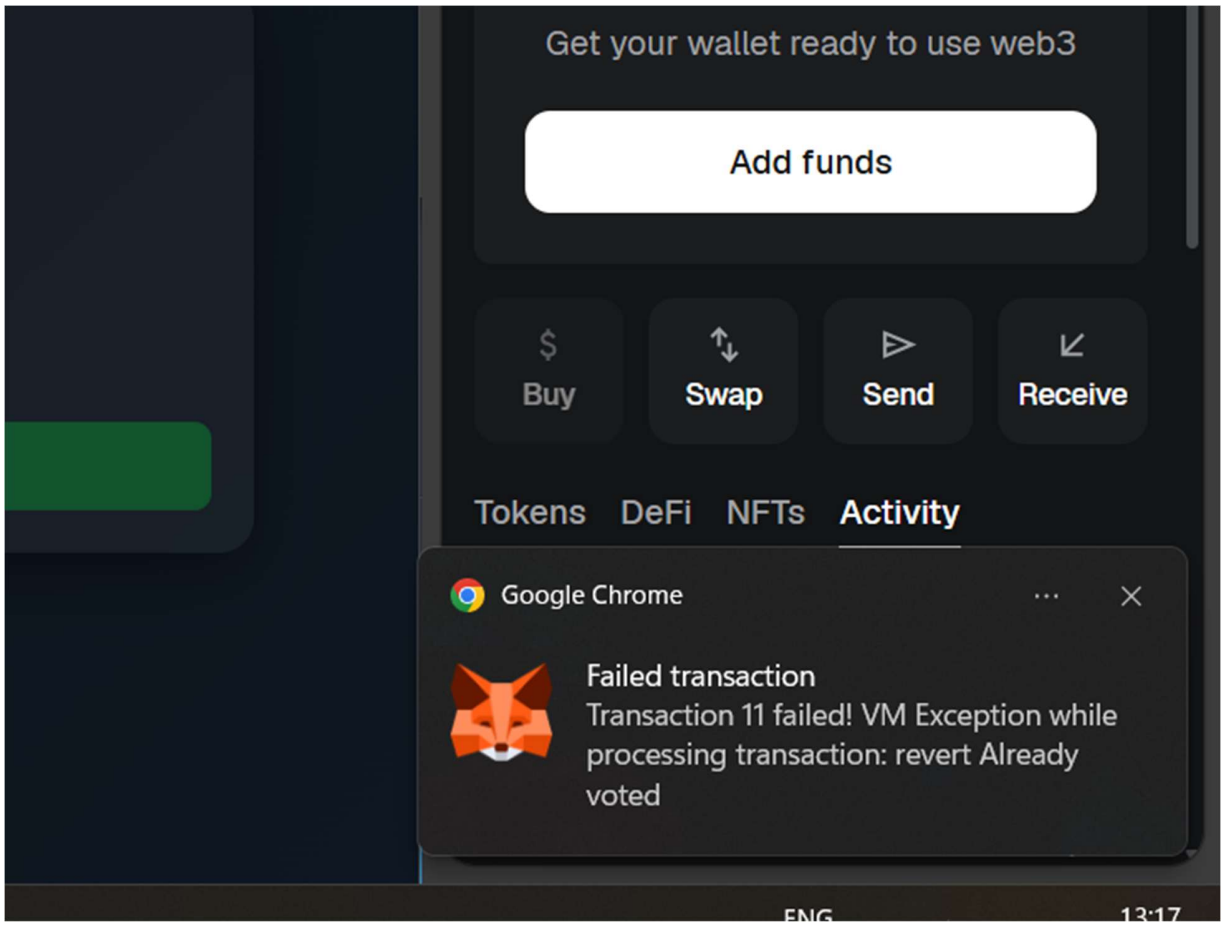
Voting Interface with MetaMask Transaction Request



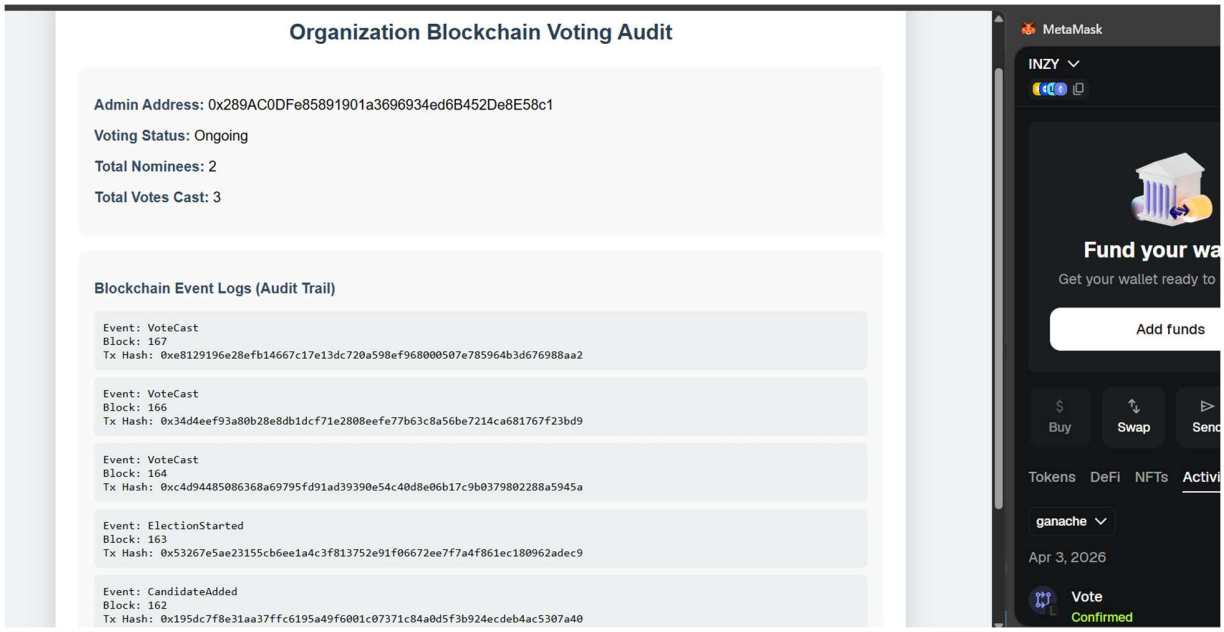
Vote Successfully Recorded on Blockchain



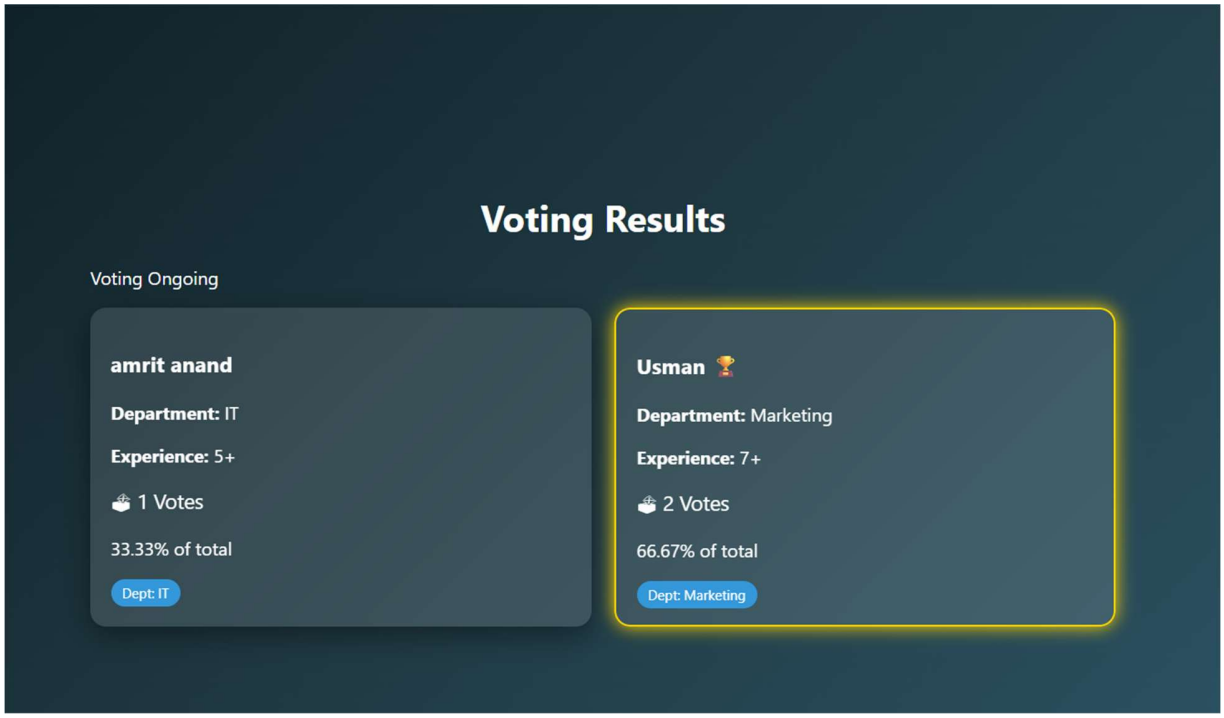
Prevention of Double Voting



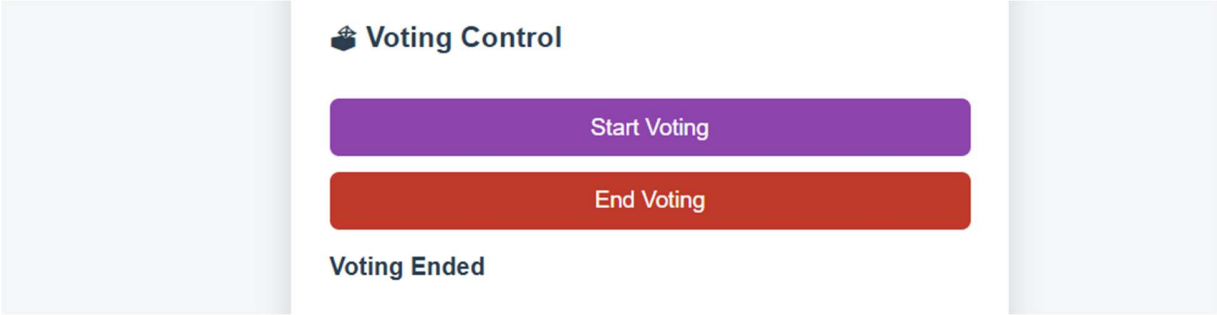
Public Audit Page with Blockchain Event Logs



Voting Results



Election Ended by Admin



Conclusion

The developed Blockchain-Based Voting System successfully demonstrates how modern technology can be used to improve the security and reliability of voting processes. Traditional voting systems often suffer from issues such as lack of transparency, chances of data manipulation, and dependency on centralized authorities. This project addresses these challenges by using blockchain technology, which ensures that all votes are stored in a decentralized and tamper-proof manner.

In this system, smart contracts play an important role in enforcing voting rules. They ensure that each user can vote only once and that votes are counted accurately without any manual intervention. Once a vote is recorded on the blockchain, it cannot be altered or deleted, which guarantees the integrity of the election process.

The implementation also provides a user-friendly interface for different roles such as admin and voters. The admin can manage candidates and control the election process, while users can securely cast their votes using MetaMask. The public audit feature further enhances transparency by allowing anyone to view election details and verify transaction logs directly from the blockchain.

Overall, the project proves that blockchain technology can provide a secure, transparent, and reliable solution for organizational voting. It reduces the risk of fraud, increases trust among users, and ensures fair election outcomes. This system

can be considered as a strong foundation for developing more advanced and scalable voting applications in the future.

Future Scope

The current system demonstrates a secure and transparent voting mechanism using blockchain technology. However, there are several areas where the system can be further improved and expanded to make it more practical and scalable.

One important area of improvement is scalability. At present, the system is designed for organizational-level usage with a limited number of users. In the future, it can be enhanced to support large-scale deployments by using more efficient blockchain networks and optimizing smart contract performance. This will allow the system to handle a large number of voters and transactions without delay.

Authentication can also be strengthened. Currently, the system verifies users through wallet-based access. This can be improved by integrating additional authentication methods such as OTP verification, employee ID validation, or biometric authentication. These enhancements will make the system more secure and reduce the chances of unauthorized access.

The system can also be extended by improving the user interface. A more user-friendly and responsive design, along with a dedicated mobile application, can make the voting process more accessible. This will encourage higher participation and make the system easier to use for non-technical users.

Another improvement is deployment on a real blockchain network. At present, the system is tested using a local environment. In the future, deploying it on a public or private blockchain network will make it suitable for real-world applications and improve its reliability.

Security enhancements can also be considered. Smart contract auditing and additional validation mechanisms can be implemented to ensure that the system is free from vulnerabilities. This will increase trust and make the system more robust.

Additional features such as real-time result analysis, notifications, and role-based dashboards can also be added. For example, users can receive updates about election status, and administrators can monitor participation levels during the voting process.

In the future, this system can be extended beyond organizations and used in universities, corporate governance, or large-scale decision-making systems. With further improvements, blockchain-based voting has the potential to become a reliable alternative to traditional voting systems.